

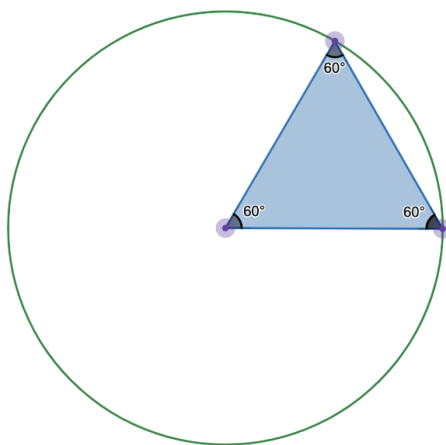
2026 Crossword Solutions

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1 1	2 3
3 5	9

ONE ACROSS

Let R be the clock's radius, θ be the angle between the two hands, and d be the distance between the clock's hands.



As illustrated in the diagram above, we're trying to find when $\theta < \frac{\pi}{3}$.

We can compare relative angular velocities to determine the probability of finding the arms in such an arrangement. Specifically, the hour hand moves at $\frac{6^\circ}{\text{min}}$ and the minute hand moves at $\frac{0.5^\circ}{\text{min}}$. Our relative speed is thus $\frac{6^\circ}{\text{min}} - \frac{0.5^\circ}{\text{min}} = \frac{5.5^\circ}{\text{min}}$. Converting 5.5° to radians, we obtain

$$\theta(t) = \frac{11\pi \text{ rad}}{360 \text{ min}} t.$$

Now we're trying to find when $\frac{11\pi}{360} t < \frac{\pi}{3} \implies t < \frac{120}{11}$.

Since time is uniformly and randomly chosen from the first 60 minutes, $\frac{120}{60} = \frac{2}{11}$. Thus $m = 2$, $n = 11$, and $m + n = \boxed{13}$.

ONE DOWN

Let $d(N)$ denote the number of positive divisors of N . Since each friend receives $d(N)$ favors and the total number of favors is 6, we have

$$(\text{number of friends}) \cdot d(N) = 6.$$

Thus, $d(N)$ must divide 6, so

$$d(N) \in \{1, 2, 3, 6\}.$$

Now check all positive integers $N < 7$:

$$\begin{aligned}
 N = 1 : d(1) &= 1 && \text{(valid)} \\
 N = 2 : d(2) &= 2 && \text{(valid)} \\
 N = 3 : d(3) &= 2 && \text{(valid)} \\
 N = 4 : d(4) &= 3 && \text{(valid)} \\
 N = 5 : d(5) &= 2 && \text{(valid)} \\
 N = 6 : d(6) &= 4 && \text{(not valid)}
 \end{aligned}$$

The valid values of N are 1, 2, 3, 4, and 5. Their sum is

$$1 + 2 + 3 + 4 + 5 = \boxed{15}.$$

TWO DOWN

We use the factorization

$$\prod_{k=0}^{2025} (x - \zeta^k) = x^{2026} - 1.$$

Substituting $x = 2078$ gives

$$\prod_{k=0}^{2025} (2078 - \zeta^k) = 2078^{2026} - 1.$$

The desired product omits the $k = 0$ term (where $\zeta^0 = 1$), so

$$N = \prod_{k=1}^{2025} (2078 - \zeta^k) = \frac{2078^{2026} - 1}{2078 - 1} = \frac{2078^{2026} - 1}{2077}.$$

We compute $N \bmod 100$. Since $2077 \equiv 77 \pmod{100}$ and $\gcd(77, 100) = 1$, we invert 77 modulo 100. Because $77 \cdot 13 = 1001 \equiv 1 \pmod{100}$, we have $77^{-1} \equiv 13 \pmod{100}$. Hence

$$N \equiv (2078^{2026} - 1) \cdot 13 \pmod{100}.$$

Reduce the base modulo 100: $2078 \equiv 78 \pmod{100}$. We compute $78^{2026} \pmod{100}$ using the Chinese remainder theorem.

Modulo 4, $78 \equiv 2$, so $2^{2026} \equiv 0 \pmod{4}$. Modulo 25, $78 \equiv 3$, and by Euler's theorem,

$$3^{2026} \equiv 3^{2026 \bmod 20} = 3^6 = 729 \equiv 4 \pmod{25}.$$

The unique number modulo 100 that is congruent to 0 modulo 4 and 4 modulo 25 is 4. Thus

$$2078^{2026} \equiv 4 \pmod{100}.$$

Therefore,

$$2078^{2026} - 1 \equiv 3 \pmod{100},$$

and

$$N \equiv 3 \cdot 13 = 39 \pmod{100}.$$

Hence, the last two digits of N are $\boxed{39}$.

THREE ACROSS

Write $2026 = 2 \cdot 1013$, where 1013 is prime. By Kummer's theorem, for a prime p , the exponent $v_p\left(\binom{n}{k}\right)$ equals the number of carries when adding k and $n-k$ in base p . Take $p = 1013$ and $n = 2026 = 2 \cdot 1013$, so $n = (2, 0)_{1013}$. Write

$k = a \cdot 1013 + b$ with $a \in \{0, 1\}$ and $0 \leq b \leq 1012$. If $b = 0$, then in the range $1 \leq k \leq 2025$ we must have $k = 1013$, and there is no carry in base 1013, hence $v_{1013} \left(\binom{2026}{1013} \right) = 0$ so $1013 \nmid \binom{2026}{1013}$. If $b > 0$, then when forming $n - k$ in base 1013 there is a borrow in the units digit, and consequently the base-1013 addition $k + (n - k) = n$ produces exactly one carry from the units place and no further carries, giving $v_{1013} \left(\binom{2026}{k} \right) = 1$. Therefore, $1013 \mid \binom{2026}{k}$ holds for every $k \in \{1, \dots, 2025\}$ except $k = 1013$, and in that case $1013^2 \nmid \binom{2026}{k}$ automatically. In particular, $1013^2 \nmid \binom{2026}{k}$ for all k , so $\binom{2026}{k}$ is never divisible by $2026^2 = 4 \cdot 1013^2$. Hence k is lucky if and only if $k \neq 1013$ and $2 \mid \binom{2026}{k}$.

It remains to count the even binomial coefficients in row 2026. By Lucas's theorem modulo 2, the number of odd coefficients among $\binom{2026}{k}$ for $0 \leq k \leq 2026$ equals $2^{s_2(2026)}$, where $s_2(2026)$ is the number of 1's in the binary expansion of 2026. Since

$$2026 = 1024 + 512 + 256 + 128 + 64 + 32 + 8 + 2,$$

we have $s_2(2026) = 8$, so there are $2^8 = 256$ odd coefficients. Thus, the number of even coefficients among $0 \leq k \leq 2026$ is $2027 - 256 = 1771$. Because $\binom{2026}{0} = \binom{2026}{2026} = 1$ are odd, the number of even coefficients among $1 \leq k \leq 2025$ is also 1771. Excluding $k = 1013$ (which is not lucky) yields $1771 - 1 = 1770$ lucky integers. $1770 = 2 \cdot 3 \cdot 5 \cdot 59$, so the answer is 59.